Subject Name: Foundation of Computer Science
Subject Code: ETCS-203
Department of CSE
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Subject: ***Discrete Mathematics***

Topic: ***Principle of Mathematical Induction***

Review

A function is defined for integers only such that $f(n) = 3^{4n} - 1$. Find;

1. $f(1) =$
2. $f(2) =$
3. $f(3) =$
4. $f(4) =$

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Did you get 80?

1. $f(1) = 80 = 80 \times 1$
2. $f(2) = 6560 = 80 \times 82$
3. $f(3) = 531440 = 80 \times 6643$
4. $f(4) = 43046720 = 80 \times 53804$

And this is an important idea. If a number is divisible by 80 then we can write it as $80 \times M$ where M is a positive integer.

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MATHEMATICAL INDUCTION !

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$$= 80 \times 1 \text{ which is divisible by } 80$$

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Step 2 : Assume true for $n = k$.

$$3^{4k} - 1 = 80 \times M$$

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$$3^{4k} - 1 = 80 \times M \quad \text{where } M \text{ is a positive integer}$$

$$3^{4k} = 80 \times M + 1$$

note that it is very important to make 3^{4k} the subject

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RTP that $3^{4(k+1)} - 1$ is divisible by 80

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$$3^{4(k+1)} - 1 = 3^{4k+4} - 1$$

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$$3^{4(k+1)} - 1 = 3^{4k+4} - 1$$

$$= 3^{4k} \times 3^{4} - 1$$

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$$3^{4(k+1)} - 1 = 3^{4k+4} - 1$$

$$= 3^{4k} \times 3^{4} - 1$$

$$= (80M + 1) \times 81 - 1$$

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RTP that $3^{4(k+1)} - 1$ is divisible by 80

$$\begin{aligned}
3^{4(k+1)} - 1 &= 3^{4k+4} - 1 \\
&= 3^{4k} \times 3^{4} - 1 \\
&= (80M + 1) \times 81 - 1 \\
&= 80M \times 81 + 81 - 1 \\
&= 80M \times 81 + 80 \\
&= 80(81M + 1)
\end{aligned}$$

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$$3^{4(k+1)} - 1 = 3^{4k+4} - 1$$

$$= 3^{4k} \times 3^4 - 1$$

$$= (80M + 1) \times 81 - 1$$

$$= 80M \times 81 + 81 - 1$$

$$= 80M \times 81 + 80$$

$$= 80(81M + 1)$$ proved by induction

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$$3^1 + 7^1 = 10$$

$$= 2 \times 5 \text{ , true for n = 1}$$

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note that it is important to make one of them the subject and it is easier to use the bigger number.

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$$3^{k+2} + 7^{k+2} = 3^k \times 3^2 + 7^k \times 7^2$$

$$= 9 \times 3^k + 49 \times 7^k$$

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$$3^{k+2} + 7^{k+2} = 3^k \times 3^2 + 7^k \times 7^2$$

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$$= 9 \times 3^k + 49\left(5M - 3^k\right)$$

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$$= 9 \times 3^k + 49 \times 5M - 49 \times 3^k$$

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$$
\begin{aligned}
3^{k+2} + 7^{k+2} &= 3^k \times 3^2 + 7^k \times 7^2 \\
&= 9 \times 3^k + 49 \times 7^k \\
&= 9 \times 3^k + 49\left(5M - 3^k\right) \\
&= 9 \times 3^k + 49 \times 5M - 49 \times 3^k \\
&= 49 \times 5M - 40 \times 3^k
\end{aligned}
$$

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3^{k+2} + 7^{k+2} &= 3^k \times 3^2 + 7^k \times 7^2 \\
&= 9 \times 3^k + 49 \times 7^k \\
&= 9 \times 3^k + 49\left(5M - 3^k\right) \\
&= 9 \times 3^k + 49 \times 5M - 49 \times 3^k \\
&= 49 \times 5M - 40 \times 3^k \\
&= 5\left(49M - 8 \times 3^k\right)
\end{aligned}
$$

jims Greater Noida

Thank You !!